

Model Selection Assignment: Cardio Risk Prediction

Problem Statement

Objective

This assignment aims to give you a practical understanding of model selection in real-world scenarios. You will learn how to analyse and preprocess data, extract meaningful features, and apply various machine learning models to uncover patterns and insights.

Through model evaluation and optimisation techniques, you will gain the ability to compare different approaches and select the most effective one. This process will enhance your skills in building reliable predictive models, improving decision making, and ensuring model accuracy and generalisation in diverse applications.

Business Value

Developing a predictive model for cardiovascular disease risk helps simplify the early stages of patient assessment. Instead of relying solely on time-consuming consultations and manual checks, a screening model can highlight individuals who may need quicker attention.

In this setting, choosing the right model becomes important: different approaches offer different strengths in terms of accuracy, interpretability, and the ability to capture complex patterns in patient data. Selecting an appropriate model ensures that high-risk cases are identified reliably while still supporting clinical decision-making.

This allows healthcare teams to prioritise patients more effectively, make better use of limited consultation time, and improve the overall flow of care.

Dataset Overview

Context

The data contains records of various patients attended to by a healthcare provider. These patients came to the healthcare provider with complaints related to cardiovascular disease.

The data records the vital statistics of each patient and whether they had the disease or not. Some of the relevant features associated with these records are the patients' height, weight, age, blood pressure, and so on.

Content

The dataset is stored as a CSV file and contains a collection of patient health data, which includes demographic information such as age, gender, height, and weight, alongside physiological measurements like blood pressure.

The data also captures various lifestyle factors, including cholesterol and glucose levels, smoking habits, alcohol consumption, and physical activity.

Instructions

1. Each learner should attempt this assignment individually.
2. Programming Language: Python
3. You will be provided with the dataset and a starter notebook. You have to perform all the tasks in the starter notebook only.
4. It is very important that you do not change any headings, subheadings, questions or tasks in your notebook as it can cause problems with grading.
5. The data will have inconsistencies and outliers please handle them as per your understanding and mention them in your report.
6. **Total Marks: 150** (100 for code notebook, and 50 for report document)

Submission Guidelines

1. To submit your solution, upload your solution files in the submission field.

You are required to upload your 2 solution files: an **Interactive Python Notebook** (.ipynb) that contains your code, and

a **Report Document** (.pdf) that presents your visualisations, analysis, results, insights, and outcomes

Please note that the solution notebook should only be generated from the starter files provided to you.

The submitted Jupyter notebook, report and PPT should contain your name and the assignment title in the format: "**CardioRisk_Prediction_MBulah**".

Mention all assumptions, if made, in the report.

The report should include the overall approach of the assignment, covering the problem statement, methodology, techniques used, and key insights and should present answers to the questions asked in the assignment, using visualisations to support your responses.

Results Expected from Learners

Briefly outline the problem statement and approach in the report document. Use visualisations in the PPT to support your answers for all questions in the assignment.

In the starter notebook, you will find headings, subheadings, and checkpoints stating the tasks you need to perform. The marks associated with each checkpoint will also be mentioned in the notebook. Keep in mind not to edit the cells with marking schemes and questions.

You can find a brief description of the tasks below.

Data Understanding [2 marks] Remove irrelevant columns [2 marks]

Data Cleaning [8 marks] Identify and handle redundant or invalid/illogical physiological values [6 marks]

Fix data types [2 marks]

EDA [27 marks] Perform univariate analysis [12 marks]

Perform correlation analysis [5 marks]

Perform bivariate analysis [10 marks]

Train-Test Split [5 marks] Define feature and target(s) variables [2 marks]

Split data into training and validation sets [3 marks]

Feature Engineering [18 marks] New feature creation [6 marks]

Combine values in categorical columns [4 marks]

Dummy variable creation [5 marks]

Feature rescaling [3 marks]

Model Building [38 marks] Build linear SVM and find optimal cutoff [18 marks]

Build simple decision tree and find optimal cutoff [12 marks]

Hyperparameter tuning [8 marks]

Predictions and model evaluation [2 marks] Make final predictions and evaluate [2 marks]

Report [50 marks] Exploratory Data Analysis and Feature Engineering [15 marks]

Model Building and Hyperparameter Tuning [15 marks]

Model cards, final insights and conclusions [20 marks]

Model Selection Assignment: Cardio Risk Prediction Assignment ID: MS/01 Total Marks: 150

Evaluation Rubrics

The following criteria will be used to evaluate your solutions.

Table 1: Rubrics Criteria	Meets expectations	Does not meet expectations
Data Understanding	1. Identifier type columns were correctly handled.	1. Identifier type columns were handled incorrectly or were not handled at all.
Data Cleaning	1. Data inconsistencies and errors were correctly identified and handled. 2. DataTypes have been corrected for columns with incorrect DataTypes.	1. Data inconsistencies and errors were not identified and handled or were handled incorrectly. 2. DataTypes have not been corrected or have been incorrectly assigned

		for columns with incorrect DataTypes.
EDA	1. Performed univariate analysis by visualising the distribution of all columns. 2. Performed correlation analysis by visualising a heatmap of the correlation matrix. 3. Perform bivariate analysis by visualising the relationship between categorical columns and the target variable.	1. Failed to plot the distributions of numerical columns or created incomplete or inaccurate plots without meaningful insights. 2. Incorrectly visualised the heatmap or failed to interpret correlations. 3. Failed to visualise the influence of categorical variables on the target variable. Provided incomplete or unclear visualisations and insights.
Train-Test Split	1. Feature and target variables are defined correctly. 2. Data is split into training and test sets maintaining the ratio of 70:30.	1. Feature and target variables are not defined or defined incorrectly. 2. Data splitting is implemented incorrectly or not implemented at all.

Feature Engineering	1. Created new feature(s) from existing features to enhance the model's ability to capture patterns. 2. Identified and combined low frequency values in categorical columns to reduce sparsity and improve model generalisation. 3. Created dummy variables for independent columns in both training and validation sets. 4. Applied feature scaling to numerical columns effectively by	1. Failed to create new feature(s) from existing features to enhance the model's ability to capture patterns. 2. Failed to identify and combine low frequency values in categorical columns to reduce sparsity and improve model generalisation. 3. Failed to identify or create appropriate dummy variables for independent columns in both training and validation sets.
----------------------------	---	--